

IMPACT OF WEATHER VARIABLES ON SHORT-TERM ELECTRICITY DEMAND FORECASTING IN SINGAPORE: A COMPARATIVE STUDY WITH PPSO-OPTIMISED MODELS



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1 Introduction

In Singapore, a compact grid serves approximately 5.9 million residents in a tropical climate that maintains the year-round cooling demand; national electricity consumption exceeded 55 TWh in 2024 (Energy Market Authority of Singapore 2025). Because air-conditioning is the largest end-use of electricity in buildings, demand is highly sensitive to small variations in temperature and humidity (Zhao and Magoulès 2012), with holidays, weekends, and the monsoon seasons adding further variability. Accurate short-term forecasting is therefore essential for reliable grid operation and cost-effective generation scheduling.

Classical methods (ARIMA, exponential smoothing, regression) assume linear, stationary relationships and struggle with the non-linear interactions among weather, calendar, and autoregressive demand dynamics (Al-Musaylh et al. 2018). Tree-based machine learning models capture these interactions well but are sensitive to hyperparameters, and manual tuning is expensive and often sub-optimal. Zhang et al. (Zhang, Y. Chen, and Yan 2023) showed that CatBoost paired with Phasor Particle Swarm Optimization (PPSO) outperforms five other meta-heuristics. But this was on hourly data from a *single residential customer*, and with tuning applied only to the favored model. Whether the approach generalizes to *daily national-level* demand in a tropical setting and whether its advantage survives when *all* competing models receive the same tuning budget remain open questions.

This project (i) builds a daily Singapore dataset merging EMA demand with Open-Meteo weather data (February 2023 – December 2025); (ii) engineers 16 features across weather, derived-comfort, calendar, and autoregressive categories; (iii) compares eight model configurations - a Lag-1 baseline, Linear Regression, Random Forest, XGBoost, and CatBoost, plus PPSO-tuned variants of each tree model under an *identical* budget - on six metrics; and (iv) deploys the best model through a web portal (FastAPI + React) for non-technical users. Intra-day resolution, sector decomposition, and deep learning models are beyond the scope of this paper.

2 Related Work

Classical time-series methods such as ARIMA and Holt-Winters handle linear temporal patterns reasonably well, but they tend to lose to machine learning once exogenous weather variables are included: for example, Almusaylh et al. (Al-Musaylh et al. 2018) showed SVR and MARS outperform ARIMA on short-term Australian demand, and Schirmer et al. (Schirmer, Mporas, and Potamitis 2019) found tree-based models clearly superior to linear regression for residential load prediction. Within the tree-ensemble family, Random Forest (Breiman 2001), XGBoost (T. Chen and Guestrin 2016; Khan, Byun, et al. 2020), and CatBoost (Prokhorenkova et al. 2018) are widely used; CatBoost’s native categorical handling is particularly useful for demand features such as day-of-week, month, and holiday flags, and weather, holidays, and weekends are well-established demand drivers (Khan, Kim, et al. 2021).

On the tuning side, meta-heuristic optimisation can make a substantial difference - Li et al. (Li et al. 2020) reported a 42% accuracy gain from PSO-tuned models on university building data. PPSO (Ghasemi et al. 2019) is a variant of PSO that uses a phase angle to modulate the cognitive and social velocity components through periodic functions, which removes the need to hand-tune inertia and acceleration coefficients. Zhang et al. (Zhang, Y. Chen, and Yan 2023) applied CatBoost-PPSO to hourly consumption of a single residential customer and found it outperformed five alternative

meta-heuristics.

However, there is a methodological concern in prior work: tuning is typically applied only to the authors’ preferred model, so it is unclear whether the reported advantage comes from the algorithm itself or simply from having better hyperparameters. Additionally, CatBoost-PPSO has not been tested on daily national-level demand in a tropical climate. Our project tackles both of these gaps.

3 Methodology

3.1 Pipeline Overview

Our system follows a modular, end-to-end pipeline (Figure 1) controlled by a single YAML configuration file. It covers data acquisition, feature engineering, a chronological train/test split, model training with optional PPSO tuning, and multi-metric evaluation. Each stage is implemented as a standalone Python module, making it straightforward to swap components or rerun individual steps.

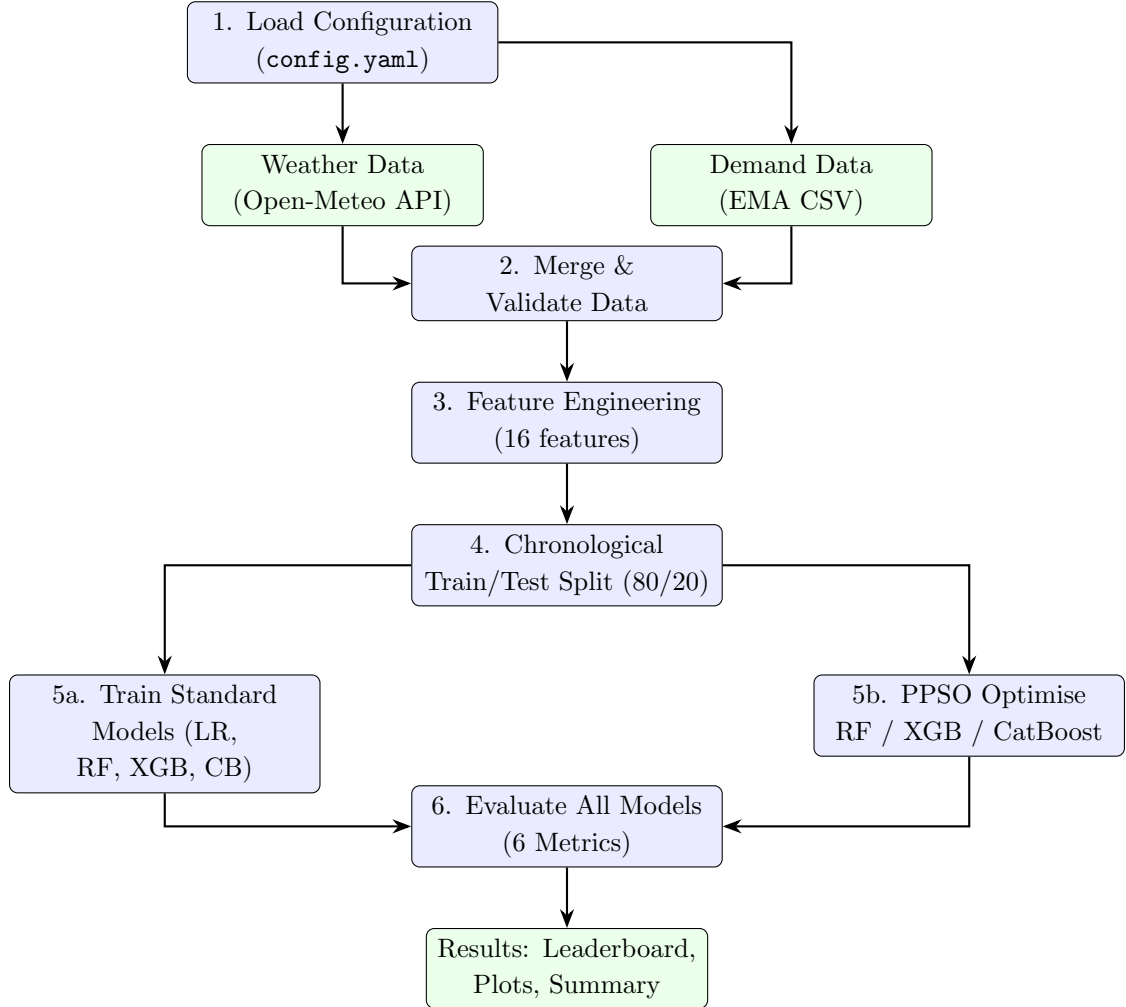


Figure 1: High-level architecture of the electricity demand forecasting pipeline.

3.2 Data and Feature Engineering

We source daily national demand (MWh) from EMA (Energy Market Authority of Singapore 2025) and retrieve four daily weather variables (mean temperature, mean relative humidity, maximum wind speed, and cumulative solar radiation) from the Open-Meteo Archive API (Open-Meteo 2025)

for Singapore’s coordinates (1.3521°N, 103.8198°E). An inner join on date gives us 1,065 observations spanning February 2023 to December 2025. Figure 2 plots the demand and weather series over this period: demand shows clear weekly cyclicality and a mild upward trend, while temperature stays within the narrow 26–30°C tropical band.

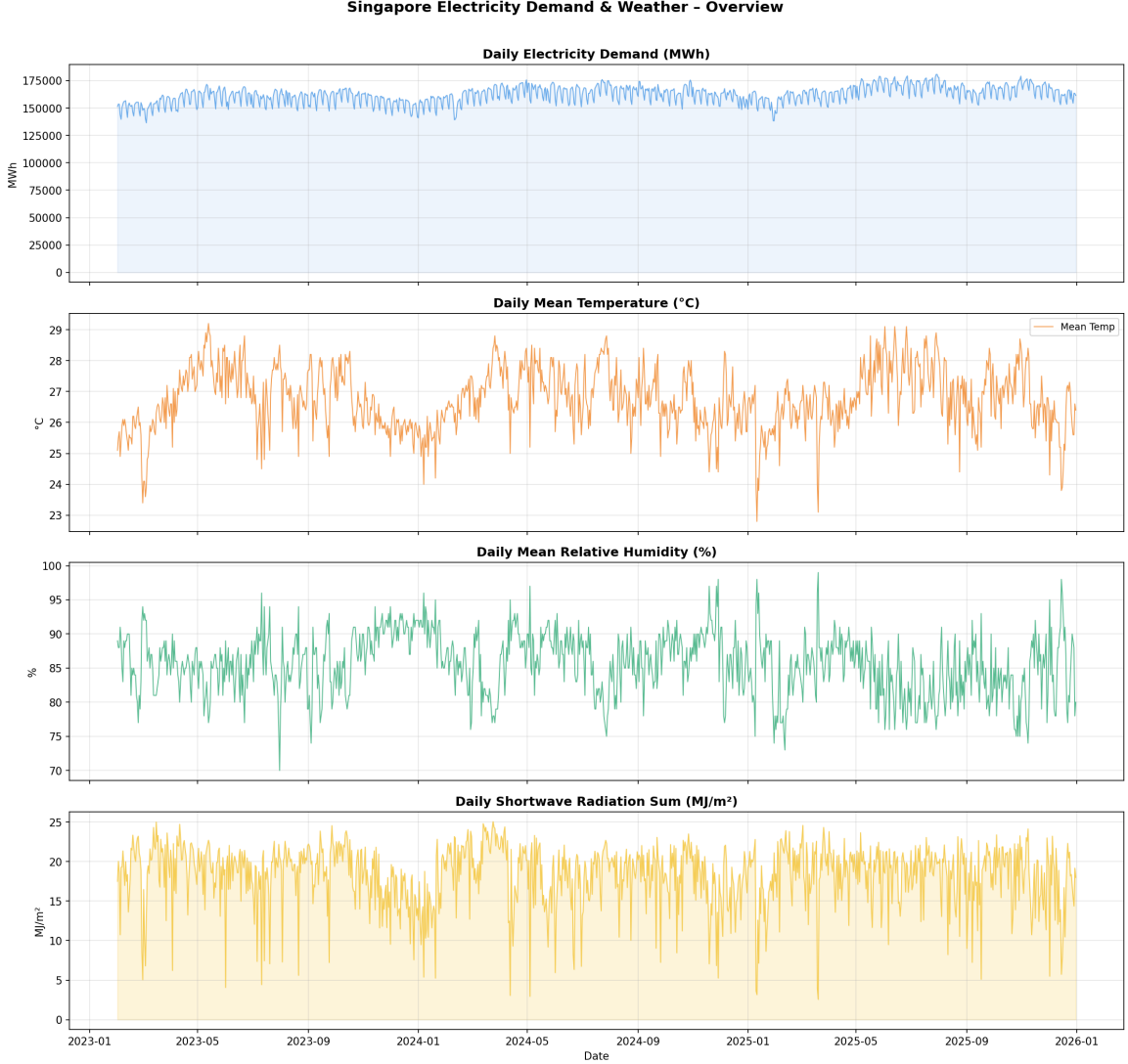


Figure 2: Daily electricity demand and weather variables in Singapore, 2023 – 2025.

Sixteen features are constructed across four categories: *weather* (the four raw variables); *derived*—a Comfort Climate Index, $CCI = T \times (1 + 0.01 \times H)$, which captures combined thermal stress, plus a season indicator; *temporal* (day-of-week, month, weekend and Singapore public-holiday flags); and *autoregressive* (demand lags at 1 and 7 days; 3- and 7-day rolling means and standard deviations). We use an 80/20 *chronological* split (852 train / 213 test samples) to avoid any forward-looking bias; features are standardised only for Linear Regression.

3.3 Models and Fair Tuning Protocol

We compare eight configurations: a Lag-1 baseline, Linear Regression, and default Random Forest, XGBoost, and CatBoost, plus PPSO-tuned variants RF-PPSO, XGB-PPSO, and CatBoost-PPSO. The key design choice here is to give PPSO to *every* tunable model under an *identical* budget, rather than reserving it for CatBoost alone as Zhang et al. (Zhang, Y. Chen, and Yan 2023) did. This way, any performance gap reflects the algorithm itself, not a tuning advantage.

We chose PPSO for three practical reasons: (i) it is essentially *parameter-free* (the phasor angle automatically balances exploration and exploitation, so we avoid the circular problem of tuning the tuner); (ii) its periodic velocity modulation handles the non-convex, expensive hyperparameter landscape well, where each fitness evaluation requires multiple full model trainings; and (iii) it was already validated against six meta-heuristics in the closest prior study (Zhang, Y. Chen, and Yan 2023). For mathematical details we refer the reader to Ghasemi et al. (Ghasemi et al. 2019).

Each model’s four-dimensional search space covers its most sensitive hyperparameters (Table 1). We use a shared budget of 15 particles $\times$ 20 iterations with early stopping after 5 stagnant iterations; fitness is the mean RMSE over 3-fold `TimeSeriesSplit` cross-validation on the training set only (Figure 3). All seeds are fixed to 42.

Table 1: PPSO hyperparameter search spaces (identical budget for all models).

Model	Tuned hyperparameters (bounds)			
Random Forest	trees (100–800)	depth (4–25)	min split (2–20)	min leaf (1–10)
XGBoost	trees (100–1000)	depth (3–10)	learn. rate (0.01–0.2)	L_2 (1–20)
CatBoost	iterations (100–800)	depth (4–10)	learn. rate (0.01–0.2)	L_2 (1–20)

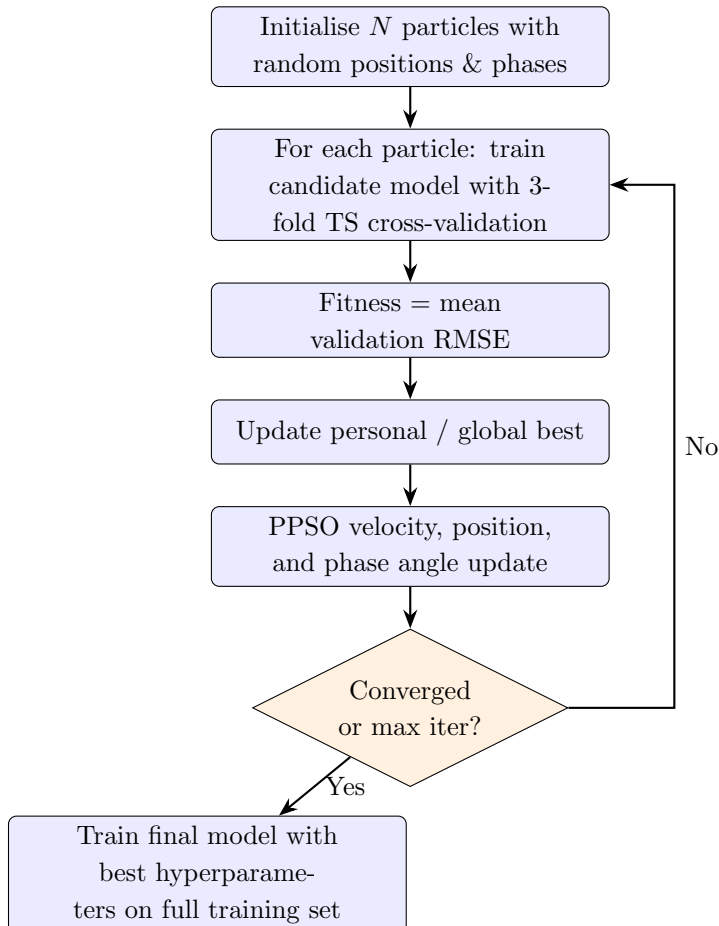


Figure 3: PPSO hyperparameter optimisation workflow.

3.4 Evaluation Metrics

We follow Zhang et al. (Zhang, Y. Chen, and Yan 2023) and report six metrics on the held-out test set: MAE, RMSE, MAPE, and RAE (all error-based; lower is better), plus R^2 and the Willmott

Index (WI) (Willmott 1982) (agreement-based; higher is better). We include WI because it is more robust to outliers than R^2 .

3.5 Web Portal

To make the model accessible beyond the research pipeline, we deploy it through a lightweight web portal. A FastAPI backend loads the serialised model and exposes REST endpoints on local-host (`/api/summary`, `/api/range`, `/api/predict`, `/api/forecast`), while a React/Vite frontend provides two views: a *History Report* page for retrospective accuracy over a user-selected period (Figure 4), and a *Demand Trends* page that produces forward outlooks using live Open-Meteo weather forecasts.

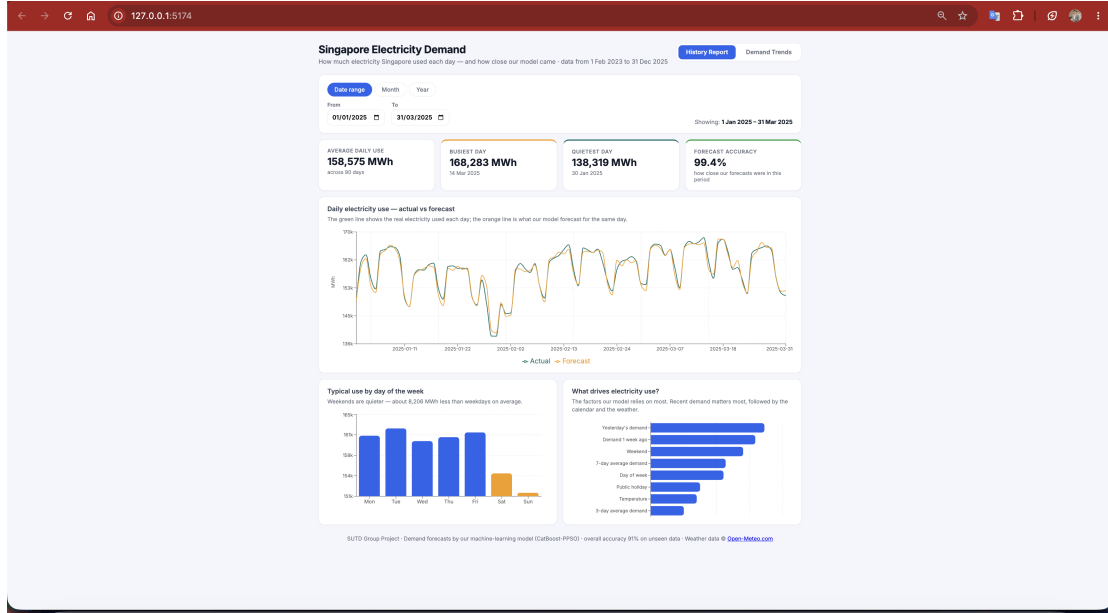


Figure 4: History Report page of the web portal: actual vs. predicted demand and summary statistics for a user-selected period.

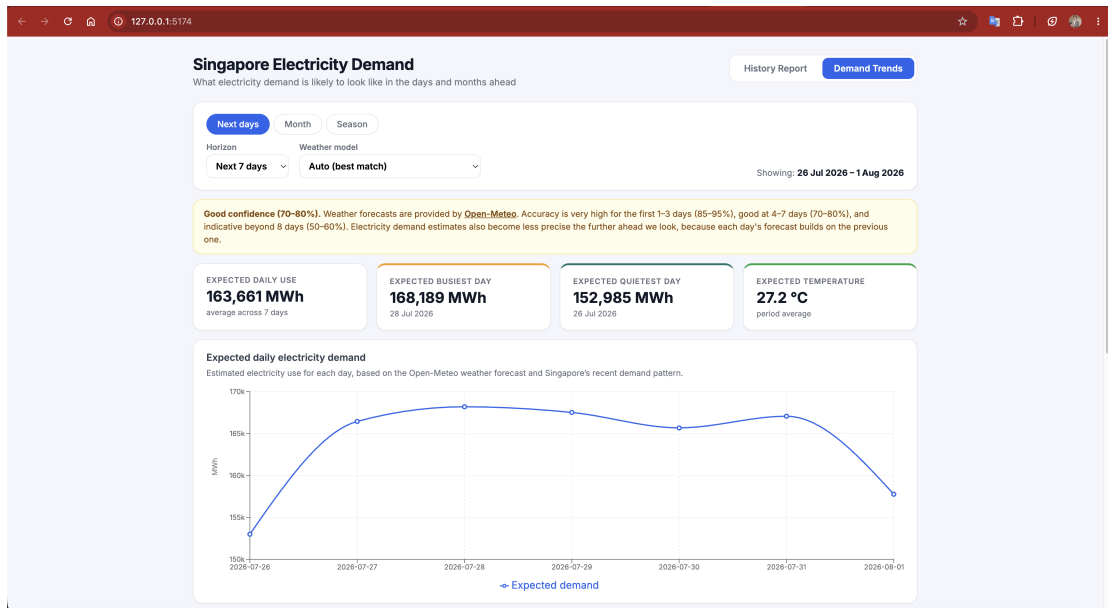


Figure 5: Demand Trend page of the web portal: showing the trend of electric demand through input time period.

4 Results

4.1 Model Comparison

Table 2 summarises test-set performance for all eight configurations, sorted by RMSE. CatBoost-PPSO clearly separates itself from the rest, while the remaining models cluster relatively tightly, and notably, Linear Regression sits among the tree-based methods rather than trailing behind them.

Table 2: Model comparison on the test set (213 samples).

Model	MAE	RMSE	R^2	MAPE (%)	RAE	WI
CatBoost-PPSO	1,496.2	1,895.5	0.9135	0.89	0.280	0.9774
CatBoost	1,679.4	2,118.8	0.8919	0.99	0.314	0.9709
Linear Reg.	1,693.4	2,119.2	0.8919	1.01	0.317	0.9731
XGB-PPSO	1,687.8	2,165.6	0.8871	1.00	0.315	0.9697
Random Forest	1,774.0	2,237.6	0.8795	1.05	0.332	0.9653
RF-PPSO	1,794.1	2,283.9	0.8744	1.06	0.335	0.9634
XGBoost	1,833.4	2,313.4	0.8712	1.08	0.343	0.9652

Key observations:

1. **CatBoost-PPSO leads on every metric:** $R^2 = 0.914$ (over 91% of demand variance explained), RMSE of 1,895.5 MWh ($\approx 1.1\%$ of mean daily demand), and MAPE of 0.89%.
2. **PPSO gains are model-dependent.** Tuning improves CatBoost substantially (RMSE -10.5%) and XGBoost moderately (-6.4%), but it proved detrimental to Random Forest ($+2.1\%$). This is probably because RF’s bagging already regularizes it well, so PPSO has less to gain, and here it looks like the optimizer mildly overfit the cross-validation folds. Since every model received the same budget, these differences confirm that CatBoost-PPSO’s lead is genuine, not a by-product of preferential tuning.
3. **Linear Regression is surprisingly competitive**, essentially tied with default CatBoost on RMSE. This tells us that the engineered autoregressive features (lag-1, rolling means) already linearise much of the problem; the value of tree models lies in capturing residual interactions between weather and calendar features.
4. **All models exceed $R^2 = 0.87$ and $WI = 0.96$** , which confirms that the 16-feature set captures the dominant demand drivers regardless of model choice.

4.2 Prediction Behaviour

Figure 6 plots actual versus predicted demand for CatBoost-PPSO across the 213-day test period. The model tracks weekly cyclicality well (weekend dips are clearly reproduced), follows demand level shifts, and tends to err most on sharp holiday-related drops, events that are hard to anticipate from weather and calendar features alone.

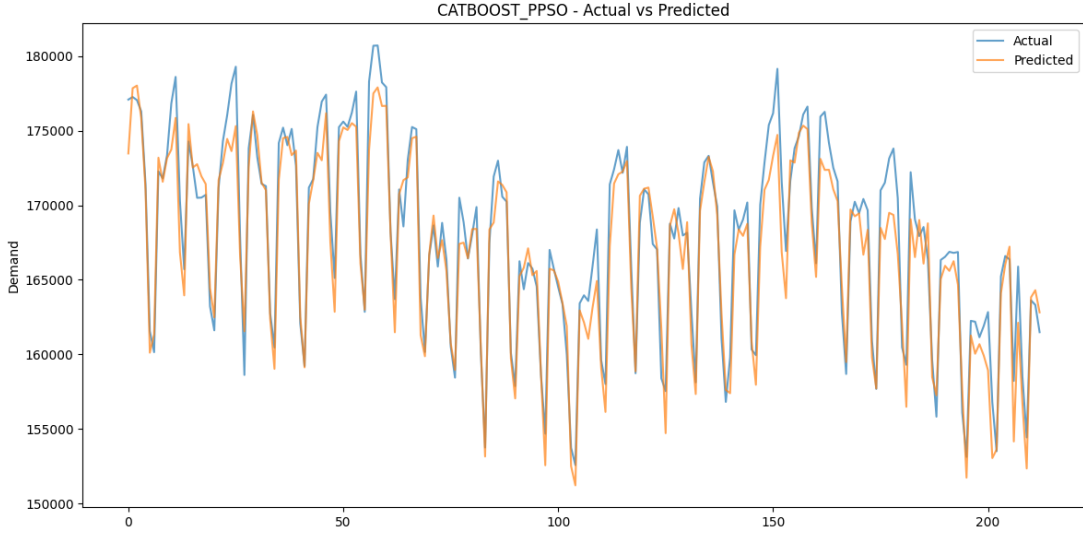


Figure 6: Actual vs. predicted daily electricity demand (MWh), CatBoost-PPSO, test set.

4.3 Feature Importance

Figure 7 shows the CatBoost-PPSO importance scores. As expected, autoregressive features dominate - demand_lag_1 and the rolling means carry the most weight - followed by day_of_week. Weather variables (temperature, humidity, CCI) sit in the mid-ranks, providing meaningful exogenous signal on top of the demand history. The same ranking holds for RF and XGBoost, suggesting this hierarchy is robust across model families.

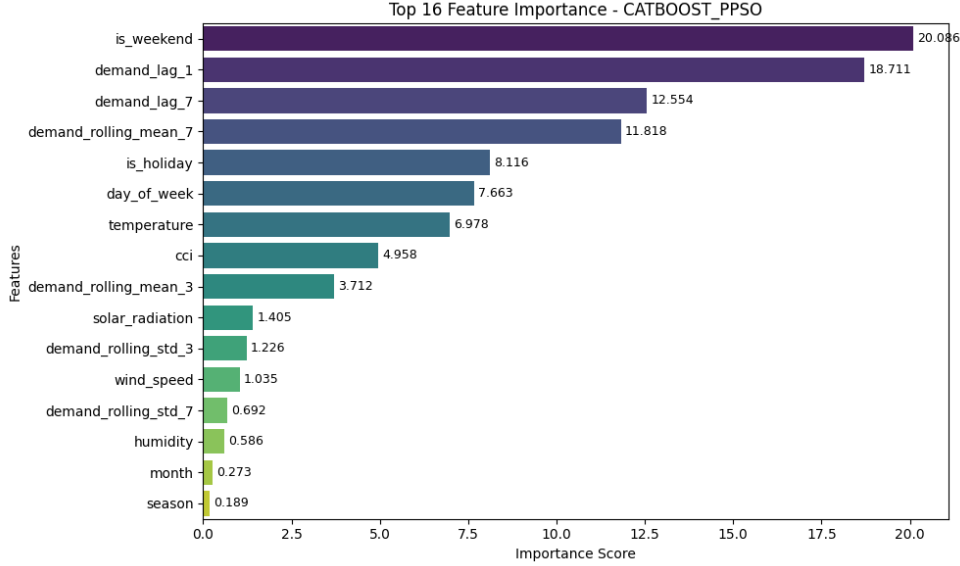


Figure 7: Feature importance, CatBoost-PPSO. Autoregressive features dominate, with weather variables providing secondary signal.

4.4 Impact of PPSO Optimisation

To isolate the tuning effect, Table 3 breaks down the before-and-after metrics for CatBoost, the model that benefited most.

Table 3: Impact of PPSO hyperparameter optimisation on CatBoost.

Metric	CatBoost (default)	CatBoost-PPSO	Improvement
MAE (MWh)	1,679.4	1,496.2	−10.9%
RMSE (MWh)	2,118.8	1,895.5	−10.5%
R^2	0.8919	0.9135	+2.2 pp
MAPE (%)	0.99	0.89	−10.6%
RAE	0.314	0.280	−10.9%
WI	0.971	0.977	+0.6 pp

Compared with Zhang et al. (Zhang, Y. Chen, and Yan 2023), who reported $R^2 = 0.971$ on hourly single-customer data, our lower R^2 (0.914) is not surprising: daily national demand aggregates consumption across millions of heterogeneous users, which inevitably adds unexplained variance. Still, the fact that the methodology transfers across such different scales, resolutions, and climates is encouraging.

5 Discussion

5.1 Interpretation

Beyond the raw rankings, three findings stand out. First, hyperparameter optimisation is not a universal win. PPSO helped the two gradient-boosted models, whose accuracy depends heavily on learning rate and regularisation strength, but it actually degraded Random Forest’s performance. This makes sense: RF’s bagging mechanism is inherently stable across a wide range of hyperparameters, so an aggressive search is more likely to overfit the cross-validation folds than to find a genuinely better configuration. Had we only tuned CatBoost, as is common in the literature, this distinction would have been invisible, the reported gains could easily have been mistaken for algorithmic superiority rather than a tuning artefact.

Second, the strong showing of Linear Regression tells us something important about the problem structure. Once we include lag-1 demand and rolling statistics, most of the daily signal is already captured linearly. The non-linear models earn their keep on the residual: weather–calendar interactions (e.g. a hot weekday versus a cool public holiday) that a linear model cannot represent. CatBoost’s native categorical handling is especially well-suited here, which helps explain why it pulls ahead of XGBoost and RF once PPSO finds good hyperparameters.

Third, using time-series cross-validation inside the PPSO fitness function turned out to be essential. Early experiments with random K-fold produced optimistic fitness values that did not hold up on the test period. Respecting temporal order in every evaluation step - from the train/test split down to the inner CV folds - was critical for honest results.

5.2 Limitations

Several limitations should be kept in mind. Daily aggregation hides intra-day peaks; Open-Meteo reanalysis data may differ from ground-station measurements; external shocks such as economic events or policy changes are not modelled; PPSO is stochastic and our results reflect a single seeded run; and multi-day portal forecasts inherit whatever error exists in the upstream weather forecast.

5.3 Future work

Moving to hourly resolution with intra-day features would be the most impactful next step. We would also like to run multi-seed PPSO experiments with statistical significance testing, benchmark sequence models (LSTM, Transformer) under the same fair-tuning protocol, and add online retraining to the deployed portal so the model stays current as demand patterns evolve.

6 Conclusion

We built an end-to-end pipeline for daily electricity demand forecasting in Singapore, training and comparing eight model configurations on a 16-feature dataset covering weather, calendar, and autoregressive variables from February 2023 to December 2025. Our main takeaways are:

1. **CatBoost-PPSO is the strongest model**, with $R^2 = 0.914$, MAE of 1,496.2 MWh, RMSE of 1,895.5 MWh, and MAPE of 0.89% on the held-out test set.
2. **Fair tuning matters.** Giving every tree model the same PPSO budget revealed that the gain is large for CatBoost (-10.5% RMSE), moderate for XGBoost (-6.4%), and absent for Random Forest. This means CatBoost-PPSO’s lead comes from the algorithm itself, not from receiving preferential optimisation.
3. **Autoregressive features carry most of the signal**; weather variables (temperature, humidity, CCI) add meaningful secondary information, consistent with Singapore’s cooling-driven demand profile.
4. **The approach generalises:** CatBoost-PPSO, originally validated on hourly single-customer data (Zhang, Y. Chen, and Yan 2023), transfers well to daily national-level demand in an equatorial climate.
5. **From pipeline to product:** we deployed the best model in a web portal that offers retrospective accuracy reports and weather-driven demand outlooks, showing a viable path from research code to a decision-support tool for grid operators and energy planners.

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